Drakkar Ascent - requirement-led results

Service-load shell screen: Aluminium 7075-T6, 1.404759 kg shell only. Required safety factors, ring/connection mass and manufacturing details remain unresolved. This is not a final passing competition design.

Parameter	Brief moment	Separate M + V*L case
Metal	Aluminium 7075-T6	Aluminium 7075-T6
Length	1500 mm	1500 mm
Outside diameter	115 mm	133 mm
Nominal wall	0.93 mm	1.06 mm
Shell mass	1.404759 kg	1.851952 kg
Assembly mass	Unresolved	Unresolved

Requirements and methods retained

Supplied loads/materials, rigid circular end boundaries and K = 1. Von Mises for metals; a declared quadratic screen for effective QI materials; Euler/beam-column mechanics; NASA SP-8007 geometric shell knockdowns. Bending remains in the shell and real joints. A knockdown factor is not an assigned safety factor.

Unsupported numerical choices removed

No baseline 1.5 load multiplier, 5% wall loss, 1.5 mm initial bow, 0.35 shell cap or 0.35 Brazier multiplier. No unpressurised case, assumed bolt/flange dimensions, invented composite ply thickness or material-specific minimum wall. Nominal thickness is used directly. The old 2.878 kg assembly used different assumptions and cannot be compared as an assembly weight saving.

Results and remaining limits

All seven materials use a common shell-only mass boundary and numerical thickness grid. Composite interaction and laminate data remain incomplete. The full report includes both moment cases, all-material load scenarios and separate closed-end pressure strength checks.

Pressure enters strength at the stated 0.3 MPa. Pressure-stabilised buckling is unresolved; the model retains a labelled no-credit capacity screen. Long-shell extrapolation, peak-shear surrogate and ideal ovalisation remain analytical limitations. Passing a numerical screen does not establish required safety-factor compliance or a global optimum.

What makes the study complete

Specify the required safety factors, design and account for the end connections/rings, and establish manufacturing/laminate details. See the full report for equations, applicability limits and sources. No FEA or physical testing was performed.